



NORTHERN UNIVERSITY

OF BUSINESS & TECHNOLOGY KHULNA

Team: CSE 4104-7A-02

Bug Tracking & Defect Resolution Report

Project Title: AI Sales Analytics Dashboard

Course Name: Software Development III

Course Code: CSE 4104

Submitted to,

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1. Defect Tracking Matrix

Bug ID	Module	Description	Severity	Status	Resolution
BUG-01	Auth	Empty login fields submitted without client/server validation	Medium	Fixed	Added Pydantic schema validation & client-side helper text alerts.
BUG-02	RBAC	Viewer role was able to post sale records	High	Fixed	Enforced <code>require_manager_or_admin</code> dependency on <code>POST /api/sales</code> .
BUG-03	RBAC	Manager role could permanently delete sales records	High	Fixed	Restricted <code>DELETE /api/sales/{id}</code> strictly to <code>require_admin</code> role.
BUG-04	AI Layer	External Gemini API 503 high-demand spike caused crash	High	Fixed	Integrated seamless offline fallback to <code>GroundedAIEngine</code> (zero downtime).
BUG-05	AI Chat	Adversarial SQL injection strings in chatbot input	Critical	Fixed	Implemented deterministic regex router + parameterized ORM queries.
BUG-06	UI/UX	Mobile sidebar drawer did not auto-collapse on navigation	Low	Fixed	Added route change event listener in Sidebar to close drawer on <code>< 768px</code> .
BUG-07	Sales UI	Fast keystrokes in search bar sent repeated redundant requests	Low	Fixed	Added 300ms debounce timer to search input handler in React.

2. Evidence of Bug Identification & Resolution

POST

/api/auth/login Login

Cancel

Reset

Parameters

No parameters

Request body required

application/json

Edit Value | Schema

```
{
  "email": "pi@dexample.com",
  "password": "admin123"
}
```

Execute

Clear

Responses

Responses

Curl

```
curl -X 'POST' \
  http://127.0.0.1:8080/api/auth/login \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
    "email": "pi@dexample.com",
    "password": "admin123"
  }'
```

Request URL

http://127.0.0.1:8080/api/auth/login

Server response

Code

Details

200

Response body

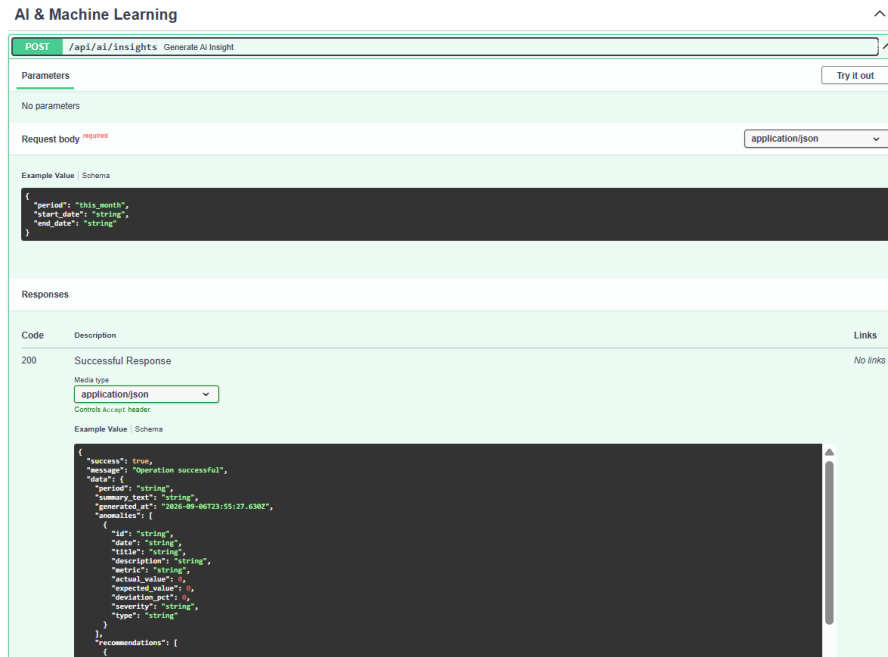
```
{
  "success": true,
  "message": "Login successful",
  "data": {
    "access_token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJ2ZW5kaW10IjyWBIzIn3v6GU1O1JhZG1pb1IsIiwV44C10MT400gyNDh0PCwiYW91ZjYwN7sg",
    "token_type": "bearer",
    "user": {
      "name": "Mohaimen Islam Pisl",
      "email": "pi@dexample.com",
      "role": "admin",
      "id": 21,
      "created_at": "2026-01-01T10:00:00"
    }
  }
}
```

Download

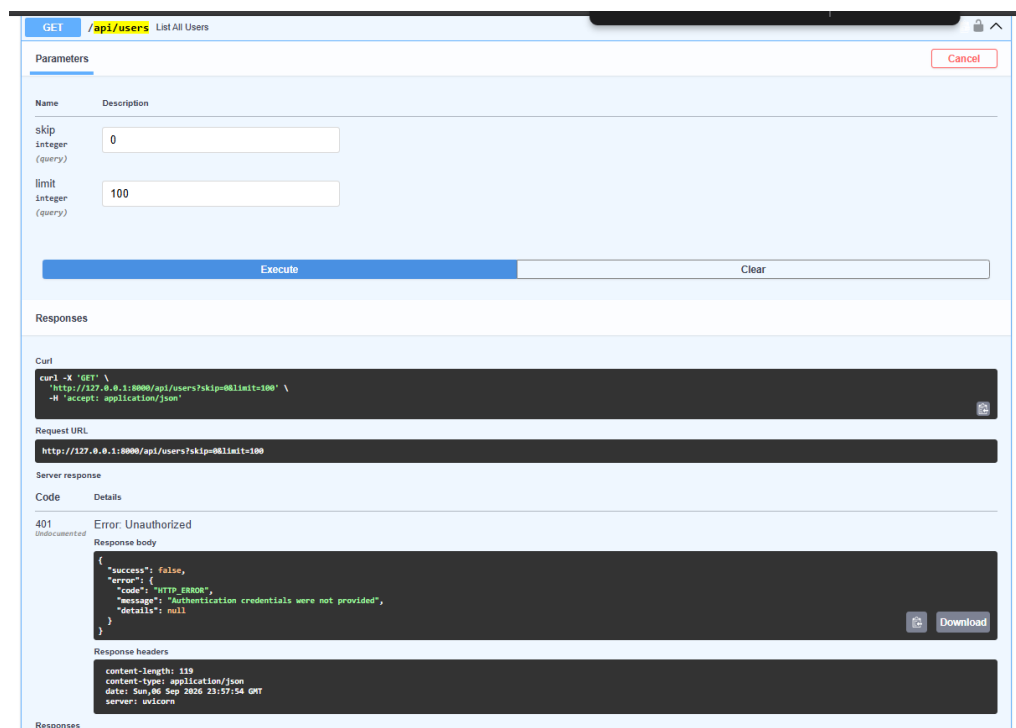
Response headers

```
access-control-allow-credentials: true
content-length: 381
content-type: application/json
date: Sun, 06 Sep 2026 23:48:18 GMT
server: uvicorn
```

Screenshot 1: Login Authorization bug fixed



Screenshot 2: AI Responsiveness checking and fixing



Screenshot 3: User sales add and remove bug fixing